

VOLTRELAY ENERGY

Network Service Quality & Performance Analysis

January 2024 – June 2025

Business Analytics Report

1. Executive Summary

This report analyzes 3,877,013 swap attempts recorded across VoltRelay Energy's six-city network from January 2024 through June 2025. The analysis focuses on reliability, customer abandonment, city and infrastructure variation, queue waiting time, and revenue.

There were 3,645,809 completed swaps, 145,959 failed attempts, 68,533 abandoned attempts and 16,712 cancelled attempts. The overall observed failure rate was 3.76%, abandonment rate was 1.77%, and recorded revenue was ₹233,126,055.55 (about ₹23.31 crore).

The strongest operational signals are failure-rate spikes in May–June 2024 and May–June 2025, higher observed failure rates in Jaipur/Delhi NCR/Hyderabad, higher observed failure for Gen1 chargers, and a strong observed association between longer queue waits and abandonment.

2. Network KPIs

KPI	Value
Total attempts	3,877,013
Completed swaps	3,645,809
Failed attempts	145,959
Abandoned	68,533
Cancelled	16,712
Failure rate	3.76%
Abandonment rate	1.77%
Revenue	₹233,126,055.55

3. Problem Understanding

The objective is to understand where and when the swap network experiences service-quality and operational issues and to identify evidence-backed areas for investigation. The analysis is descriptive/diagnostic because the dataset does not define a single predictive target.

- How do failure and abandonment rates change over time?
- Which cities and stations show higher observed failure rates?
- How does charger generation relate to observed failures?
- How does queue waiting time relate to abandonment?
- Which operational areas should be prioritized for investigation?

4. Data Understanding

The organizer-provided data includes swap events, stations, riders, batteries, station hourly status, support tickets, city context and fleet partners. The primary swap_events dataset contains 3,877,013 rows and 22 columns, covering 2024-01-01 00:00:33 to 2025-06-30 23:59:23.

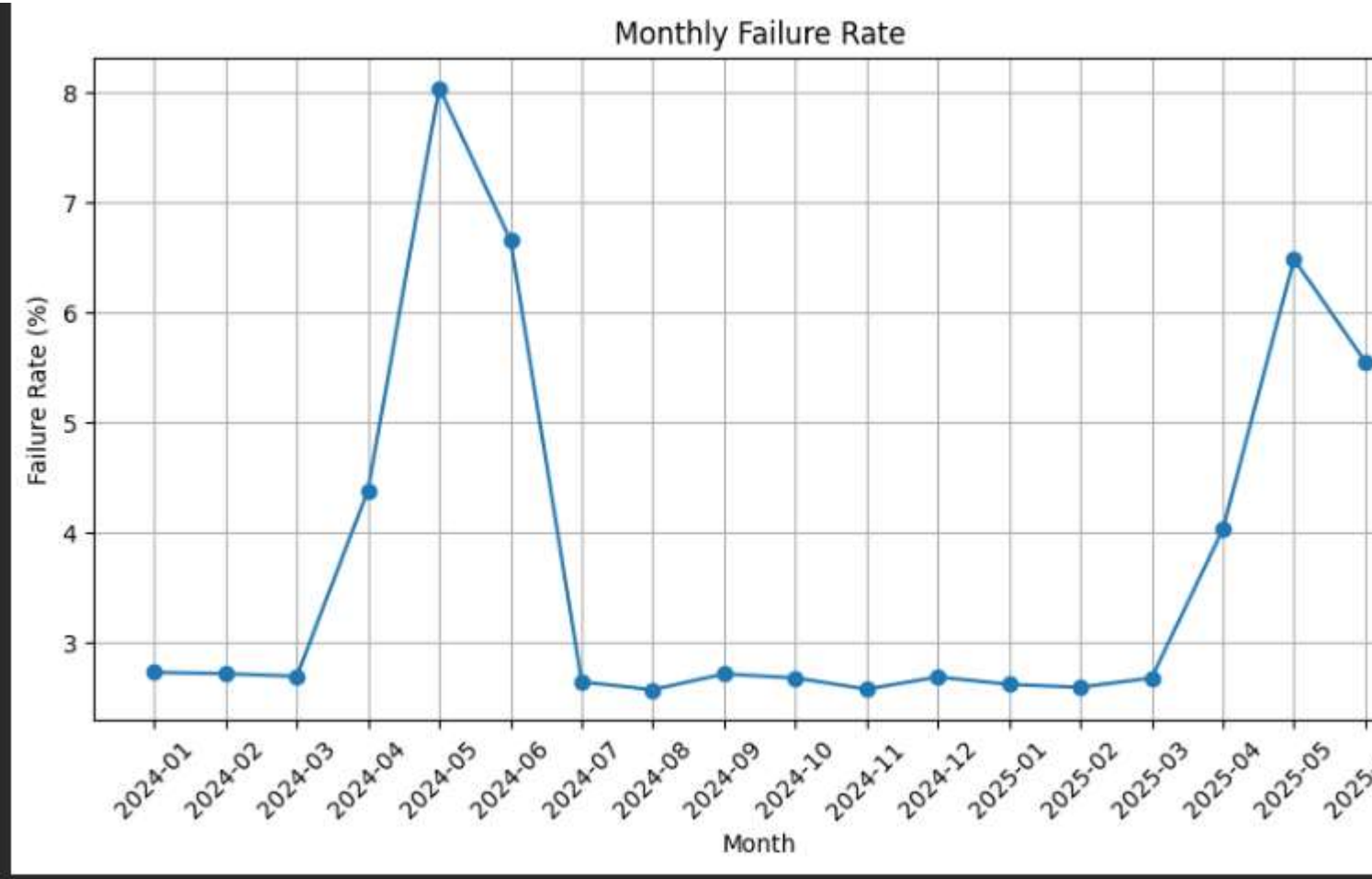
Dataset	Rows	Role
swap_events.csv	3,877,013	Primary transaction and outcome data
stations.csv	152	City, zone and charger attributes
riders.csv	20,000	Rider and vehicle attributes
batteries.csv	6,500	Battery lifecycle/health
station_hourly_status.csv	1,487,712	Hourly operational status
support_tickets.csv	44,000	Customer support
city_daily_context.csv	3,282	Weather/grid/context
fleet_partners.csv	12	Partner information

5. Data Cleaning & Validation

- Original rows: 3,877,013; cleaned rows: 3,877,013; removed rows: 0.
- Duplicate event IDs: 0.
- Missing/invalid event_ts rows: 0.
- The complete event period is January 2024 to June 2025.
- Battery, SOC and energy fields are event-type dependent; missing values for incomplete/failed events were retained rather than blindly imputed.
- Station city and charger-generation attributes were joined using station_id for infrastructure analysis.

6. Monthly Performance

Failure rates were generally around 2.6–2.7% in stable periods, with major spikes in May–June 2024 and April–June 2025. Revenue generally increased as transaction volume expanded.

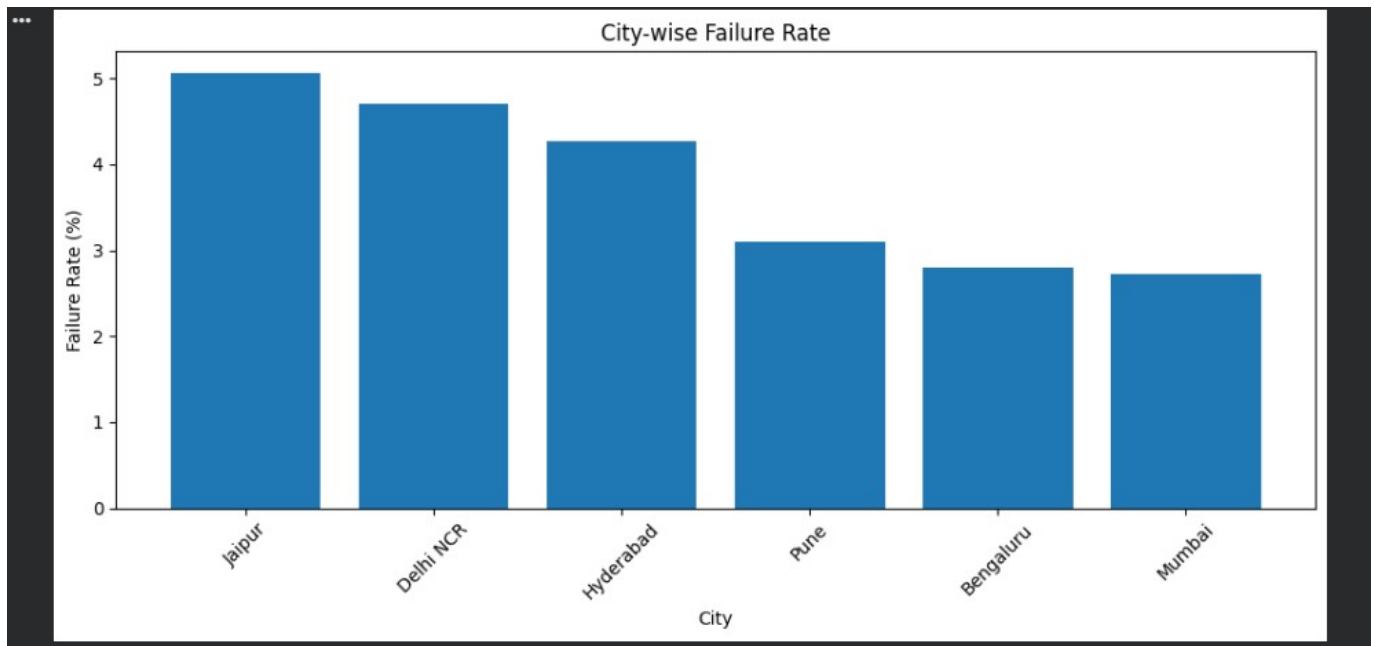


Month	Attempts	Completed	Failed	Abandoned	Revenue	Failure	Abandonment
2024-01	110348	105490	3008	1405	₹6,318,995	2.73%	1.27%
2024-02	111531	106582	3027	1451	₹6,387,615	2.71%	1.30%
2024-03	123900	118444	3332	1575	₹7,093,617	2.69%	1.27%
2024-04	131818	122836	5775	2683	₹7,370,256	4.38%	2.04%
2024-05	146278	128328	11760	5520	₹7,689,594	8.04%	3.77%
2024-06	148888	133743	9911	4596	₹8,003,276	6.66%	3.09%
2024-07	158688	151824	4186	2025	₹9,879,525	2.64%	1.28%

2024-08	172295	164931	4424	2185	₹10,724,297	2.57%	1.27%
2024-09	195885	187302	5311	2442	₹12,115,210	2.71%	1.25%
2024-10	229901	219838	6148	2904	₹14,577,272	2.67%	1.26%
2024-11	238352	228253	6131	2959	₹14,759,656	2.57%	1.24%
2024-12	264265	252621	7091	3371	₹16,343,647	2.68%	1.28%
2025-01	281957	269843	7376	3591	₹17,451,452	2.62%	1.27%
2025-02	268169	256672	6943	3386	₹16,607,262	2.59%	1.26%
2025-03	304163	290872	8136	3824	₹18,826,219	2.67%	1.26%
2025-04	312845	293089	12589	5754	₹19,004,534	4.02%	1.84%
2025-05	341733	307882	22157	10243	₹20,006,538	6.48%	3.00%
2025-06	335997	307259	18654	8619	₹19,967,092	5.55%	2.57%

7. City-wise Failure Analysis

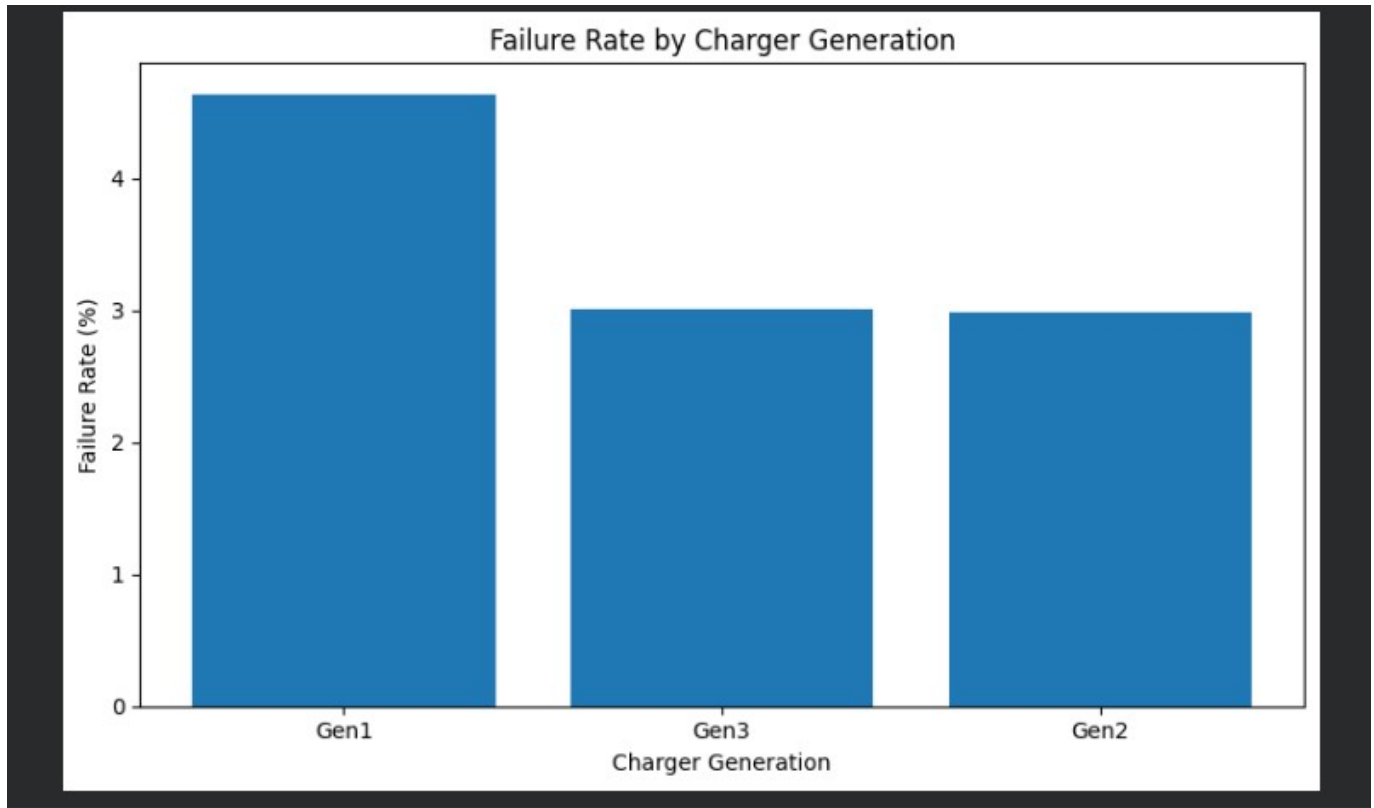
Observed failure rates differ across cities: Jaipur 5.06%, Delhi NCR 4.71%, Hyderabad 4.27%, Pune 3.10%, Bengaluru 2.80% and Mumbai 2.72%. These are observed differences and do not establish that city itself causes failures.



City	Attempts	Failed	Failure Rate
Jaipur	487,634	24,677	5.06%
Delhi NCR	768,913	36,197	4.71%
Hyderabad	711,930	30,427	4.27%
Pune	552,910	17,127	3.10%
Bengaluru	825,327	23,090	2.80%
Mumbai	530,299	14,441	2.72%

8. Charger Generation

Gen1 had the highest observed failure rate at 4.64%, compared with 3.01% for Gen3 and 2.99% for Gen2. The pattern should be investigated together with city, station age, demand and other operating conditions rather than interpreted as causal.



Generation	Attempts	Failed	Failure Rate
Gen1	1,822,034	84,499	4.64%
Gen3	372,241	11,192	3.01%
Gen2	1,682,738	50,268	2.99%

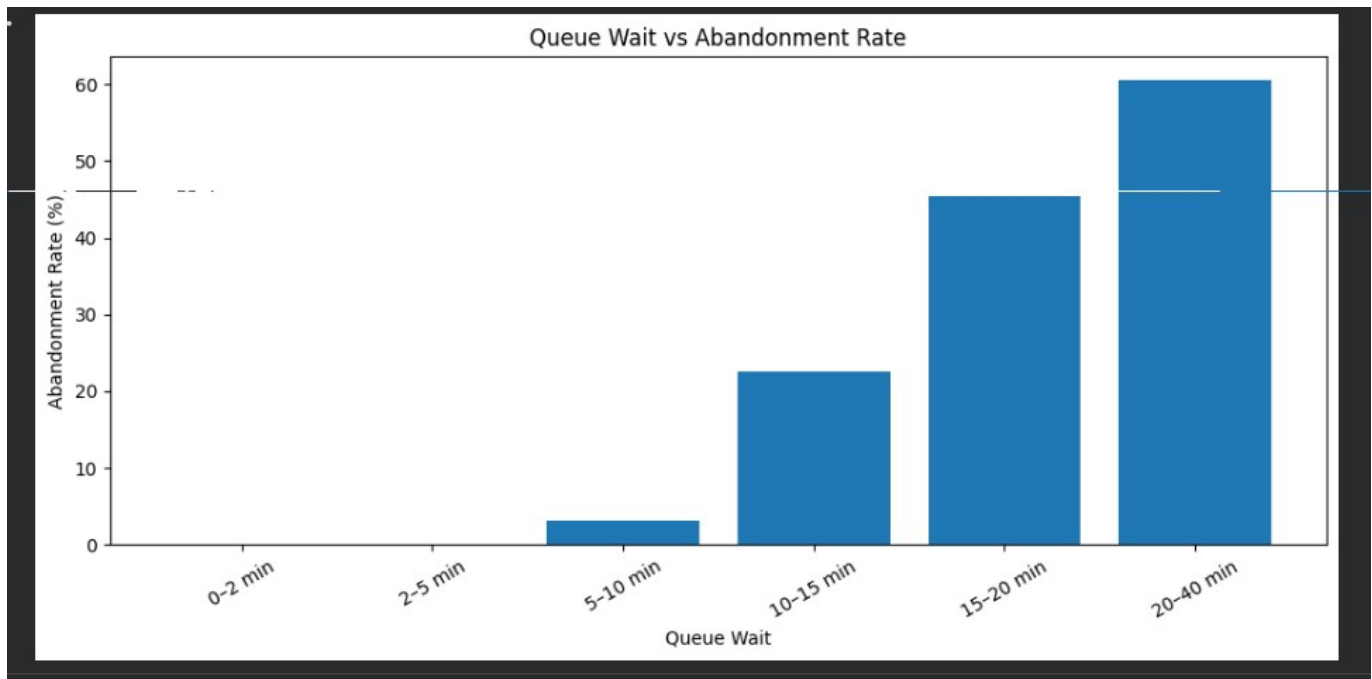
9. Failure Type Analysis

No-charged-battery failures were much more frequent than system errors: 134,389 versus 11,570 events.

City	Failed: No Charged Battery	Failed: System Error
Bengaluru	20,602	2,488
Delhi NCR	33,892	2,305
Hyderabad	28,274	2,153
Jaipur	23,303	1,374
Mumbai	12,838	1,603
Pune	15,480	1,647

10. Queue Wait & Abandonment

Queue waiting time shows the strongest observed association in the analysis. Abandonment rises sharply as waits become longer, especially after 10 minutes.



Wait bucket	Attempts	Abandoned	Abandonment Rate
0–2 min	738,654	0	0.00%
2–5 min	2,137,467	1,855	0.09%
5–10 min	859,729	27,132	3.16%
10–15 min	112,369	25,383	22.59%
15–20 min	21,719	9,877	45.48%
20–40 min	7,075	4,286	60.58%

The observed abandonment rate rises from 3.16% at 5–10 minutes to 22.59% at 10–15 minutes, 45.48% at 15–20 minutes and 60.58% at 20–40 minutes. This is an association, not proof of causation.

11. Hourly & Station Diagnostics

Average hourly wait time remains relatively stable at roughly 242–245 seconds, while evening hours 19:00–22:00 show higher observed failure and abandonment rates. Station-level analysis identifies high-failure locations for targeted investigation; station performance should be evaluated using failure rate, abandonment, wait time, volume and infrastructure attributes together.

12. Key Business Findings

- Failure rates showed significant spikes during May–June 2024 and May–June 2025.
- Jaipur, Delhi NCR and Hyderabad had higher observed failure rates than the other cities.
- Gen1 chargers showed the highest observed failure rate among charger generations.
- Longer queue waits were strongly associated with higher abandonment, with a sharp increase after 10 minutes.
- Revenue generally increased over the analysis period while service quality varied by period.
- No-charged-battery failures were substantially more frequent than system-error failures.

13. Business Recommendations

- 1. Prioritize high-failure stations:** Use failure rate, failure volume, queue wait and abandonment together to create a station investigation list.
- 2. Improve charged-battery availability:** Investigate inventory positioning, charging capacity and replenishment at stations with repeated no-charged-battery failures.
- 3. Reduce prolonged queue waits:** Use operational alerts as queues approach 10 minutes and prioritize actions during high-volume periods.
- 4. Investigate Gen1 stations:** Review maintenance history, station age, demand and local conditions before considering targeted upgrades.
- 5. Build continuous monitoring:** Maintain a city/station/hour dashboard covering attempts, failures, abandonment, queue wait, completed swaps and revenue.

14. Limitations

- The analysis covers swap-event data from January 2024 to June 2025.
- The analysis is observational; associations should not automatically be interpreted as causal.
- City, station, charger generation, demand and operating conditions may interact.
- Different segments can have different event volumes, so rates should be interpreted with sample size.
- Further analysis could investigate battery suppliers, support tickets, rider retention, pricing, weather/context and station-hour capacity.

15. Conclusion

The analysis of more than 3.87 million swap events shows that network performance varies across time, cities and infrastructure. The strongest operational signals are failure-rate spikes, city-level differences, higher observed failure rates for Gen1 chargers, and the strong observed association between long queue waits and customer abandonment.

The main operational priorities are targeted station investigation, charged-battery availability, queue-time reduction and continuous performance monitoring. These priorities should be validated through station-level and time-based operational follow-up before broad infrastructure decisions are made.

16. AI Use Disclosure

AI assistance was used for report structuring, wording refinement and presentation preparation. Quantitative findings are based on the organizer-provided dataset and verified notebook outputs.

SWAP ATTEMPTS	COMPLETED	FAILURE RATE	RECORDED REVENUE
3.88M	3.65M	3.76%	₹23.31 Cr